Egypt Car Accidents 🚗

Project Overview

This project analyzes car accidents in Egypt from 2022 to 2024.

The dataset is real and collected from the official CAPMAS (Central Agency for Public Mobilization and Statistics) website.

The analysis focuses on understanding injuries, deaths, and trends over time to provide insights for improving road safety.

🔗 Google Drive Link

All project files are available on Google Drive:

🔗 https://drive.google.com/drive/folders/10C9qdOvMabBRY6b3wbOYhe-dZJVNiaE7?usp=sharing

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Tools Used

- Excel: Data collection
- Power BI: Data cleaning, data modeling, and dashboard creation
- Python: Forecasting models and machine learning

Project Structure

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Egypt Car Accidents Analysis (2022–2024)
│
├── Data
│  ├── Raw Data – original PDF files from CAPMAS
│  └── Processed Data – cleaned and structured Excel files
│
├── Power BI
│  ├── Data Cleaning – Power Query steps
│  ├── Data Modeling – fact & dimension tables
│  └── Dashboards – Power BI files (.pbix)
│
├── Python
│  ├── Data Preparation – preparing data for modeling
│  ├── Forecasting Models – prediction of future accidents
│  └── Machine Learning – models and experiments
│
├── Visualization
│  ├── Dashboard Screenshots – exported dashboard images
│  └── Design Assets – icons and UI elements
│
├── Storytelling
│  └── Final Presentation – PowerPoint / PDF with insights
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Names of Team Members

1- Mohamed Nasser Abd el Basset

2- Ahmed Abdelrhman Hassan

3- Youssef Ahmed Ibrahim

4- Sara Basher Selem

5- Tasneem Reda Gomaa

6- Mariam Mohammed Abdelmalek

Each Member's Responsibility

- Team Leader & Support: Mohamed Nasser Abd El Basset
- Data Cleaning & Preprocessing: Sara Basher Selem , Mariam Mohammed Abdelmalek
- Data Modeling & DAX: Youssef Ahmed Ibrahim
- Dashboard Design: Tasneem Reda Gomaa
- Python (Forecasting & Machine Learning): Ahmed Abdelrhman Hassan
- Presentation & Storytelling: Mohamed Nasser Abd El Basset

Instructor

Instructor : ENG: Karim Bakly

⚙️ Data Processing & Transformation

The dataset was collected from the official CAPMAS website in the form of PDF reports containing multiple tables.

1. Data Extraction

- Extracted tables manually from PDF files into Excel format.
- Structured each table properly to prepare it for further processing.

2. Data Cleaning (Power BI)

- Imported the data into Power BI using Power Query.
- Handled a large number of missing values (nulls).
- Added proper column names and fixed inconsistent structures.

3. Data Organization

- Separated the data by year (2022, 2023, 2024).
- Each year contained 10 tables:
  - 5 tables for injuries
  - 5 tables for deaths

4. Data Merging

- Merged all years together:
  - Reduced 30 tables → 10 tables
- Combined injuries and deaths:
  - Reduced 10 tables → 5 tables
- Final merge:
  - Combined all tables into a single unified dataset

5. Data Modeling

- Transformed the final dataset into a data model:

- Created Fact table
- Created Dimension tables
- Built relationships between tables (Star Schema).

6. Measures & Calculations

- Created DAX measures for analysis (e.g., totals, percentages, comparisons).

7. Dashboard Development

- Designed and built interactive dashboards in Power BI.
- Used Adobe Photoshop to enhance the dashboard design.
- Collected icons from online resources to improve visualization.

8. Forecasting (Python)

- Exported the processed data for use in Python.
- Built forecasting and machine learning models to predict future trends.